DIWA
CanSat 2026

DIWA-1

An amateur atmospheric data acquisition CanSat

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THE SIGNIFICANCE OF SATELLITES

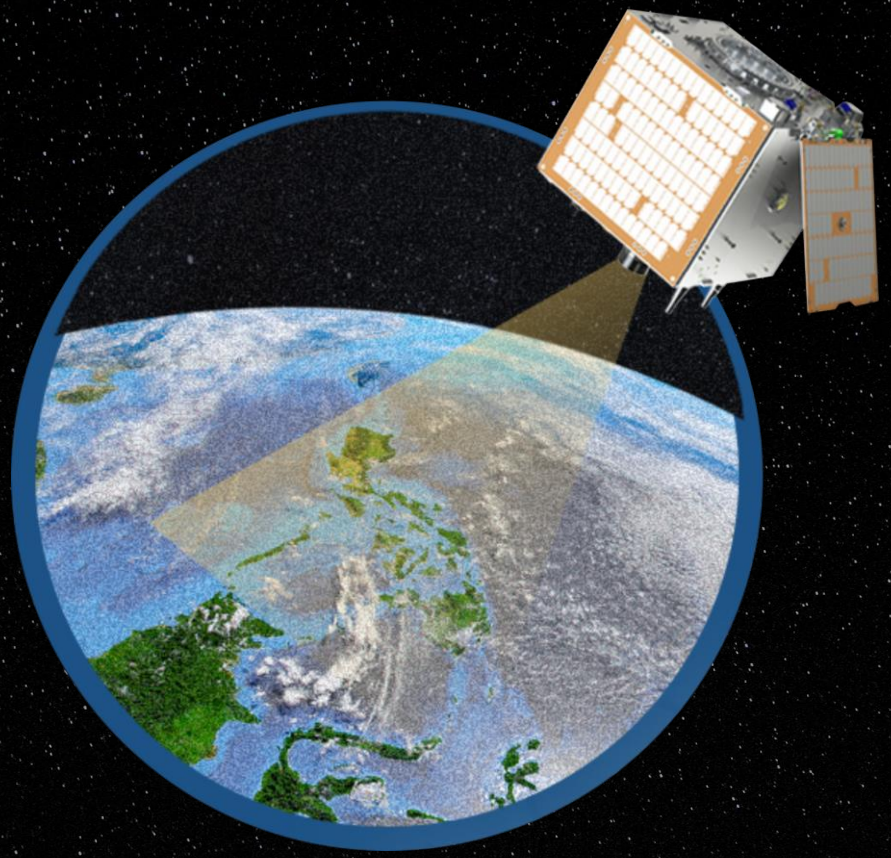


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WHY THE NAME "DIWA"?



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What is a CanSat?

OBJECTIVES

- Gather Barometric Data in hPa
- Measure Ambient temperature in °C
- Current altitude or altitude change during flight/movement

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All with a sampling rate of 1 Hz

10–SECOND STATIC MEASUREMENT TEST

DONE IN STA. MESA MANILA

- The averaged recorded temperature measurements must be within ±1–2°C of actual temperature in Sta. Mesa.

- The averaged recorded measurements should be within the range of 1005-1012 hPa (local range).

- The averaged recorded altitude measurements should be within ±3–5 m of the actual altitude of Sta. Mesa above sea level.

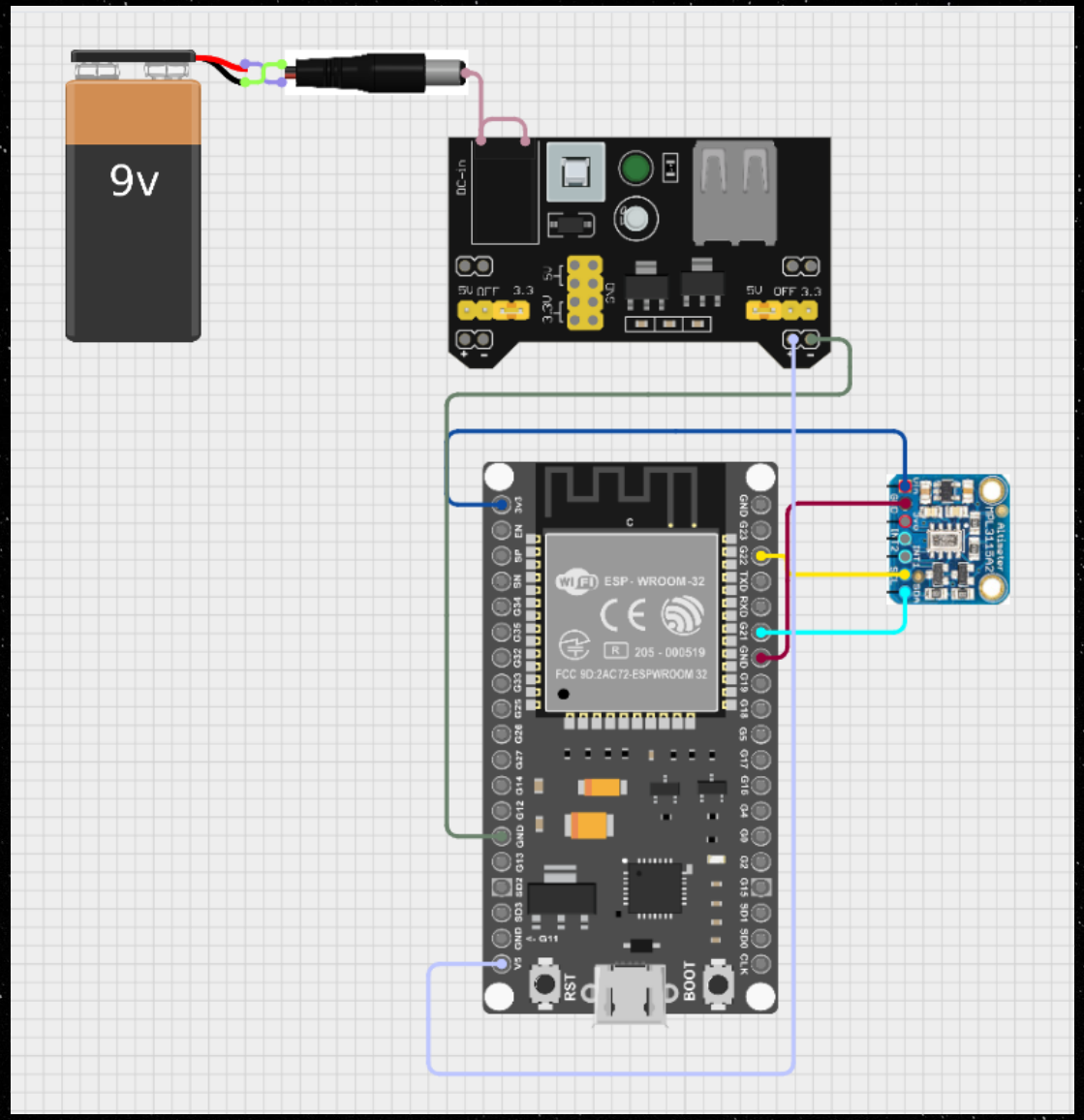
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FLOW DIAGRAM

Power is switched on -> Data transmission begins -> Microcontroller (transceiver) communicates through ESP-NOW protocol -> Transceiver receives data -> Arduino IDE displays the data received from the sensors

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SYSTEM ARCHITECTURE


9v
DC-in
5V OFF 3.3
ESP-WROOM-32
205 - 000519
FCC ID:2AC72-ESPWROOM32
RST
BOOT

ESP DEVKIT V1 38 PIN

The “brain” of the CanSat

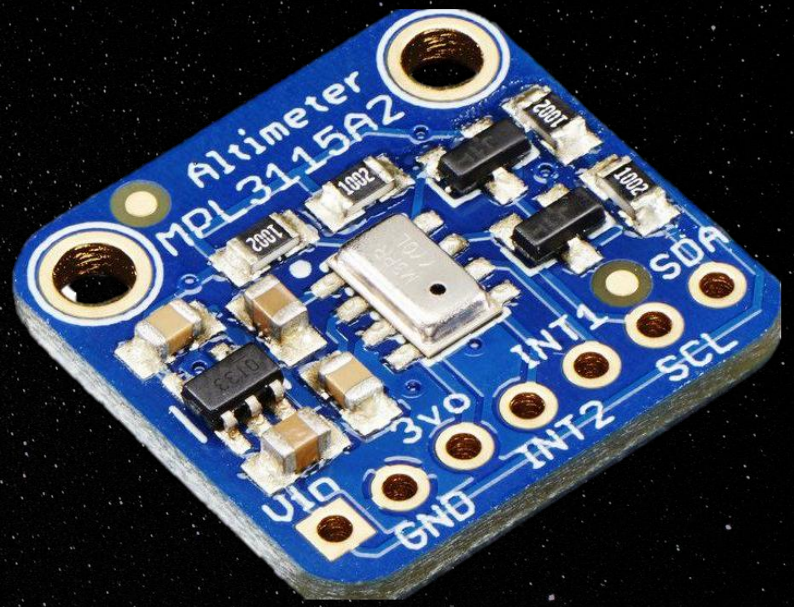
- Responsible for sending and receiving data



MPL3115A2

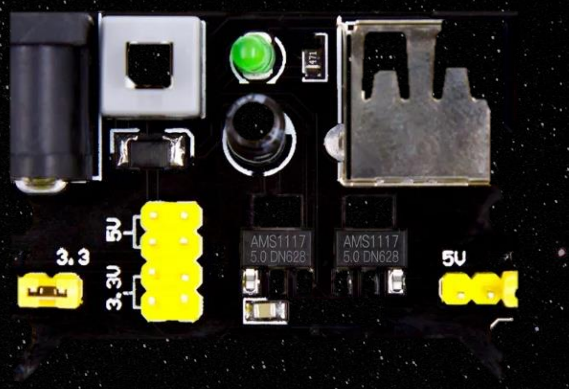
The main sensor

- Responsible for remote sensing
- Measures air pressure, altitude, temperature



POWER: 9V BATTERY -> MB102 POWER MODULE





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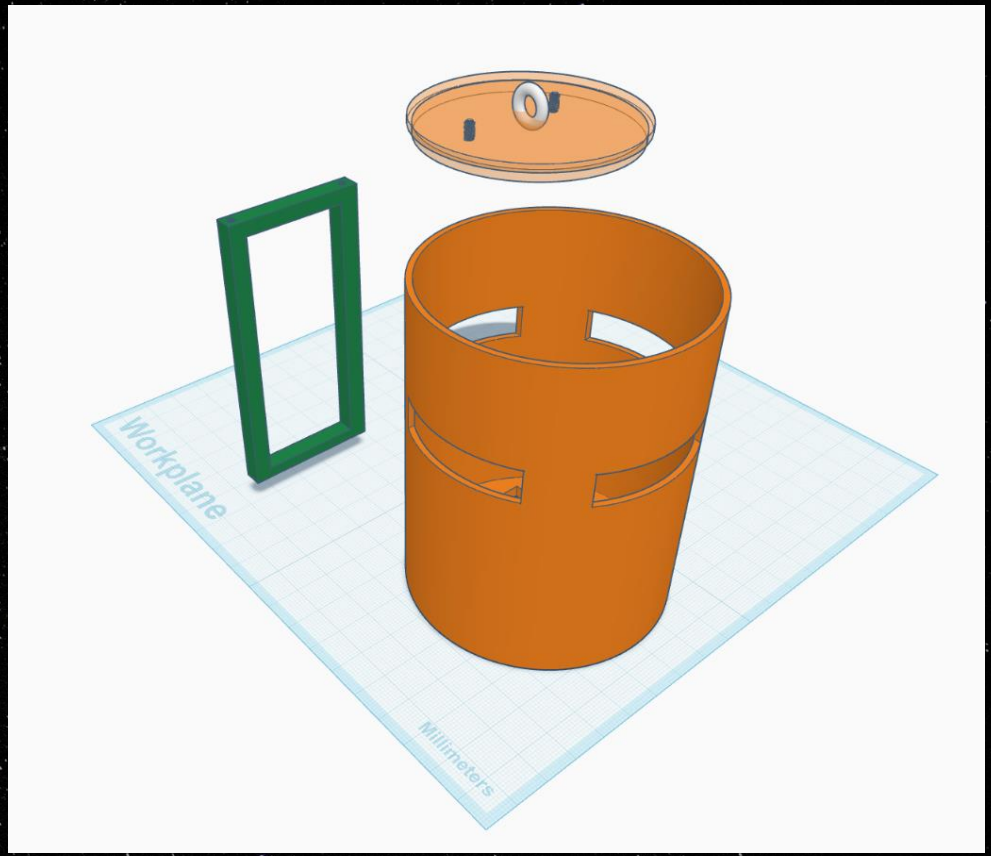
3D PRINTED STRUCTURE



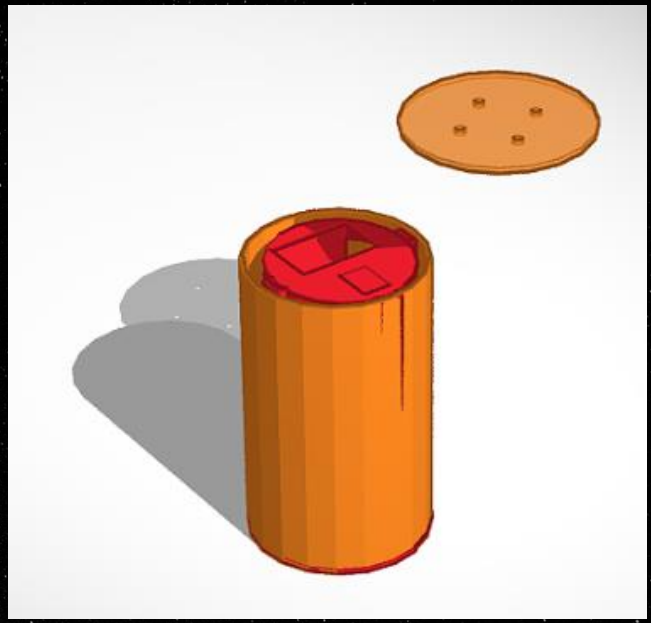
MATERIAL : Made out of PLA

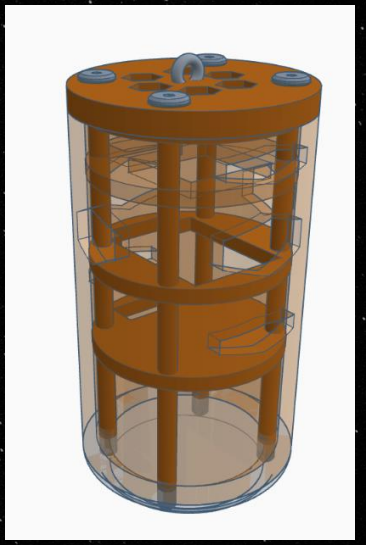
DIMENSIONS: Roughly 76 mm x 136 mm

Consists of a body, frame, and a lid



EARLIER MODELS





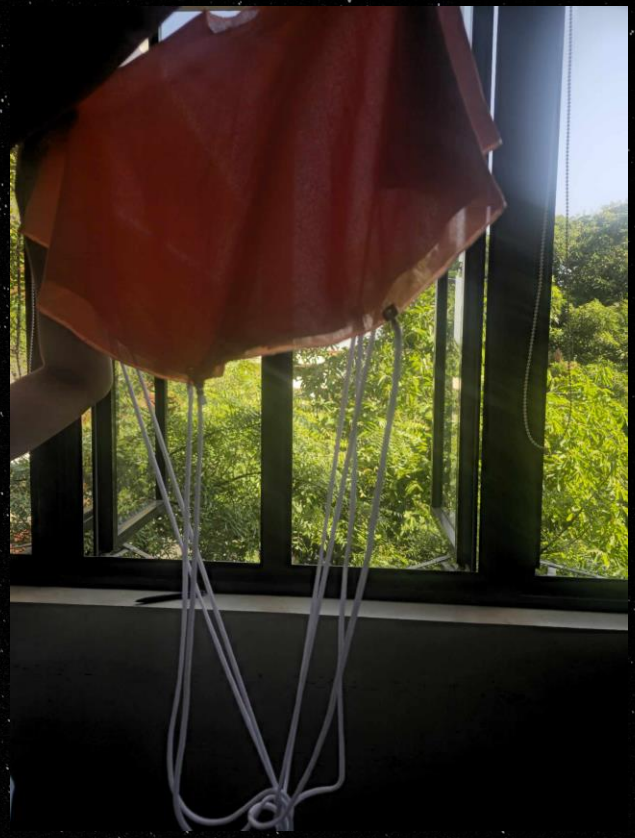
PARACHUTE SYSTEM

MATERIAL : Ripstop nylon fabric and polyester rope

DIMENSIONS:

Diameter – 0.677 m
Surface area – 0.36^2
Predicted Descent Rate: 3.6 m/s

Tied to the 3D Printed Eyebolt of DIWA-1



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SOFTWARE

```
void loop() {
  sensor barometermessagething;
  barometermessagething.pressure_inhg = baro.getPressure();
  barometermessagething.altitude_m    = baro.getAltitude();
  barometermessagething.temperature_c = baro.getTemperature();
  barometermessagething.msg_count     = msg_count++;

  Serial.printf("[Message #%lu] %.4f hPa | %.2f m | %.2f C\n",
    barometermessagething.msg_count, barometermessagething.pressure_inhg,
    barometermessagething.altitude_m, barometermessagething.temperature_c);

  if (!broadcast_peer.send_message((uint8_t *)&barometermessagething, sizeof(barometermessagething))) {
    Serial.println("Broadcast failed.");
  }

  delay(150);
```

```
  sensor barometermessagething;
  memcpy(&barometermessagething, data, sizeof(barometermessagething));

  Serial.println("---------------------------------------");
  Serial.printf("Message #%lu from " MACSTR "\n",
    barometermessagething.msg_count, MAC2STR(addr()));
  Serial.printf("  Pressure    : %.4f Hectopascal hPa\n", barometermessagething.pressure_inhg);
  Serial.printf("  Altitude    : %.2f meters\n",      barometermessagething.altitude_m);
  Serial.printf("  Temperature : %.2f *C\n",          barometermessagething.temperature_c);
  Serial.println("---------------------------------------");
}
```

STATIC TEST TABLE

The table lists 10-second readings from the sensor, with averages calculated at the bottom.

	Altitude (in m)	**Pressure** (in hPa)	**Temperature** (in °C)
1 second	10.44 meters	1010.6725 hPa	31.19 °C
2 seconds	10.44 meters	1010.6475 hPa	31.19 °C
3 seconds	10.50 meters	1010.6275 hPa	31.19 °C
4 seconds	10.50 meters	1010.6650 hPa	31.19 °C
5 seconds	10.31 meters	1010.6800 hPa	31.25 °C
6 seconds	10.25 meters	1010.6525 hPa	31.19 °C
7 seconds	10.56 meters	1010.6575 hPa	31.25 °C
8 seconds	10.38 meters	1010.6875 hPa	31.19 °C
9 seconds	10.50 meters	1010.6700 hPa	31.25 °C
10 seconds	10.25 meters	1010.6650 hPa	31.25 °C
AVERAGE	**10.413 meters**	**1010.6625 hPa**	**31.214°C**

(DIWA-1 Static Test was done in Santa Mesa, Manila)

Average elevation of Santa Mesa, Manila: 9m
Average pressure in Santa Mesa, Manila: 1005 hPa -1012 hPa
Actual Temperature at the time of testing: 31°C

VERIFICATION

The averaged sensor measurements demonstrate strong compliance with verification criteria. Temperature at 31.214°C falls well within the ±1–2°C tolerance of an actual value around 31°C, with a minimal deviation of just 0.214°C. The pressure averages 1010.6625 hPa which coordinates inside the local range of 1005–1012 hPa. Additionally, the altitude at 10.413 m aligns precisely with the ±3–5 m threshold describing the elevation of the area.

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THANK YOU !

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